Arduino-Based Smart Car Parking System Using Computer Vision

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***Abstract*— The growing number of vehicles in urban areas has led to severe challenges in parking space management, often resulting in traffic congestion, operational delays, and reliance on inefficient manual systems. The project presents an affordable smart car parking system based on the Arduino UNO microcontroller and computer vision principles to automate the process of vehicle identification and gate control. The system uses OpenCV for license plate detection and EasyOCR for optical character recognition, enabling accurate and contactless vehicle verification. Infrared (IR) sensors monitor the occupancy of parking slots in real time, and a servo motor controls gate access based on the validation outcome. An OLED display provides users with instant slot status. The system uses robust serial communication between Arduino and PC to synchronize hardware and software level events. Experimental results demonstrate license plate recognition accuracy of 91% or better and average gate actuation delay of 1.3 seconds, confirming the system's viability for efficient and scalable city parking solutions.**

***Index Terms*— Arduino UNO, Computer Vision, Number Plate Recognition, OpenCV, Smart Parking System.**

I. INTRODUCTION

A. Background of Study and Motivation

With the rapid pace of urbanization and phenomenal expansion in automobile ownership, parking management within urban environments has also emerged as a significant issue. Traditional parking systems tend to be grounded on manual observation, paper tickets, or basic RFID solutions, which are typically inefficient, error-intolerant, and non-salable. These features contribute to traffic congestion, time loss, and increased fuel consumption. As cities transition towards being smarter and more environmentally friendly, a smart parking system is an urgent necessity.

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The motivation behind the project is to design an automated, cost-effective, and reliable system using embedded systems and artificial intelligence that can facilitate the parking process.

B. Objectives

The principal objective of this project is to develop and deploy an automated parking system that tackles urban parking issues by utilizing computer vision techniques in conjunction with embedded systems technology (Arduino Uno), while considering sociological and cultural requirements.

- To design a low-cost smart car parking system using Arduino Uno, IR sensors, servo motors and computer vision to detect and manage available parking spots automatically.
- To implement a real-time computer vision-based license plate recognition system using OpenCV, YOLOv3 and EasyOCR to monitor vehicle entries and exits securely and efficiently.
- To control parking gate access based on license plate verification.
  To provide live feedback through an OLED display and LED indicators.
- To simulate and test the entire parking system in real-time conditions to validate its accuracy, reliability, and user-friendliness under various urban scenarios.
- To analyze survey data and real-world user feedback to ensure that the system is adaptable to the cultural, societal, and infrastructural contexts of Bangladeshi cities.

C. Overview of the Report

The remainder of the report is structured as follows:
Section II presents a literature review of related works in smart parking and vehicle identification. Section III outlines the methodology, system modeling, working principles, experimental setup and hardware components. Section IV presents results from both simulation and hardware testing, including cost analysis and system limitations. Section V concludes the report and discusses future improvements.

II. LITERATURE REVIEW

A. Introduction to Literature Review

Designing an effective and reliable smart parking system requires a sound understanding of modern technologies, approaches, and real-world challenges. In the last few years,

numerous developments have taken place in areas such as embedded systems, computer vision, and automatic vehicle identification. Exploring these works helps to identify the progress made so far and the existing gaps. In this chapter, recent research efforts toward smart parking systems, in particular those addressing vehicle recognition, hardware automation, and intelligent slot management, are covered.

B. Review of Relevant Works

[1] Al-Kharusi et al. (2014) used ultrasonic sensors to detect vacant parking spots, while a microcontroller system managed entry and exit. However, this system primarily focused on space detection and lacked integration with vehicle identification.

[2] Prasad et al. (2016) developed an automated system that utilized image processing techniques to recognize vehicle plates, allowing only authorized vehicles to access secured parking areas, but it a used heavy computational model.

[3] Ibrahim et al. (2013) developed an Automatic Number Plate Recognition (ANPR) system, but it relied on high-resolution cameras and heavy computational resources.

[4] Viola et al. (2001) used Haar cascades, a machine learning object detection method for their system, but it is an outdated approach.

[5] Smith (2007) developed a system that used Optical Character Recognition (OCR) using Tesseract Engine for extracting alphanumeric characters from detected plates. However, newer tools like EasyOCR have improved efficiency and accuracy.

[6] Kaur et al. (2016) demonstrated a system using Arduino to control a servo motor based on serial inputs, showcasing effective hardware-software communication. But it only focused on the car entry-exit aspects and lacked vehicle identification and automation capabilities.

C. Summary of Literature Review

While numerous smart parking systems exist, many are either cost-prohibitive or rely on sophisticated infrastructure (e.g., RFID, IoT cloud-based systems). Limited studies have combined low-cost microcontroller platforms with real-time number plate recognition for localized and affordable smart parking solutions.

Thus, developing a Car Parking System using Arduino UNO and basic computer vision techniques addresses the gap by creating a cost-effective, scalable, and easily deployable solution for small to medium parking lots.

III. METHODOLOGY AND MODELING

A. Introduction

The methodology of this project combines embedded system design with computer vision to develop a smart car parking solution capable of automating vehicle entry using license plate recognition and space availability detection. The system architecture integrates an Arduino UNO microcontroller, servo motors, IR sensors, a camera module, and computer vision software based on OpenCV and EasyOCR.

To better understand the parking-related challenges and assess the demand for such automated systems among the academic community, a preliminary survey was conducted among students. A structured questionnaire was distributed, and approximately 40 responses were collected. The survey focused on the participants' experiences with traditional parking systems, perceived inefficiencies, and their openness to using automated solutions. The results of this survey provided valuable insights that influenced key system features, such as automation, contactless operation, and user-friendly feedback.

This section presents the design and development methodology for the proposed smart car parking system. It includes the working principle, step-by-step process of development, a detailed description of all components used, and the test setup configuration for validating system performance.

B. Working Principle of the proposed project

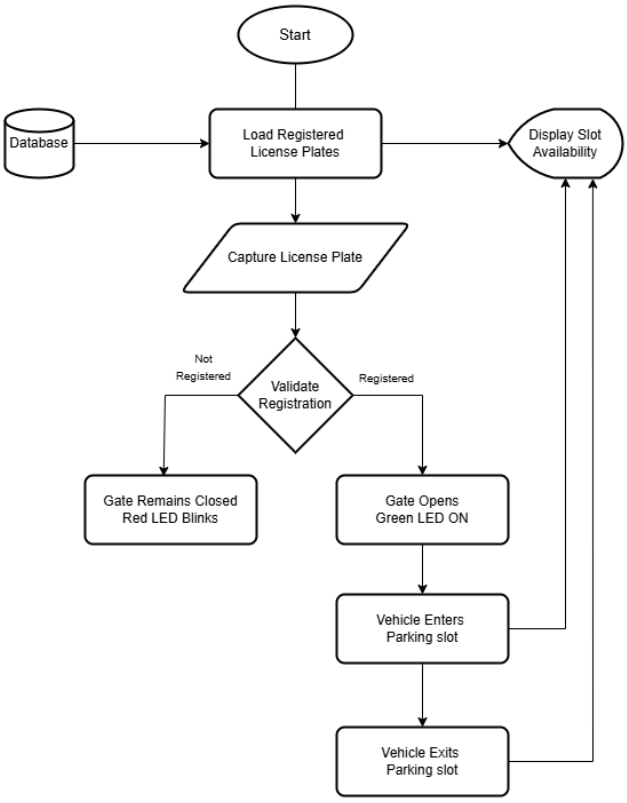


Fig. 1. Process Flowchart

The system functions by detecting a vehicle's license plate using a camera, which is processed in real-time using a Python script running OpenCV and EasyOCR for plate detection and recognition. The plate is detected 30 times, and the most frequent plate is accepted for accuracy. If the recognized plate matches any entry in the pre-registered list, a "valid" signal is sent to the Arduino via serial communication. The Arduino then activates a servo motor to open the gate and a green LED at the gate indicates validation. If the plate is not recognized, an "invalid" signal is sent, and the gate remains closed with red LED blinking as feedback. The IR sensors at each parking slot detect vehicle presence. Slot availability is updated in real-time and shown in the OLED display. Each vehicle is registered for a specific slot. If a slot is empty the

red LED at the slot is turned on and the OLED display shows "Empty" for the specified slot. When the registered vehicle enters the parking lot and is situated at the designated parking slot the green LED at the designated parking slot is turned on and the license plate is shown with the slot on the OLED display. When a vehicle leaves their slot and arrives at the gate the IR sensor detects the vehicle and opens the gate.

C. Process of Work (Experimental and Simulation)

The development process followed a structured and iterative workflow, integrating both software simulation and physical hardware testing. The major phases included:

1. Computer Vision and Plate Recognition Development:
The initial phase involved implementing a number plate detection and recognition module using Python. OpenCV and a pre-trained YOLOv3 model were used for localizing license plates in real-time video frames. EasyOCR was then applied to extract text from the detected plates. The system was tested on various sample images and live video feeds to optimize recognition accuracy under different lighting and angle conditions.

2. Arduino-Based Control Logic Design:
Simultaneously, the Arduino UNO was programmed to interpret commands received via serial communication from the PC. The logic controlled a servo motor for gate operation, IR sensors for detecting vehicle presence in entry and slots, and LEDs/OLED for feedback. Multiple input-output scenarios were simulated and validated.

3. System Integration and Communication Testing:
Once both subsystems were stable, they were integrated. A Python script on the PC continuously communicated with the Arduino to verify plates, trigger gate movement, and update slot statuses. Handshaking mechanisms and response timings were fine-tuned to ensure seamless real-time operation.

4. Experimental Testing and Validation:
The fully assembled system was tested using mock vehicles and license plates. Various entry scenarios—including valid, invalid, and re-entry—were executed. Parameters such as gate delay, sensor accuracy, display response, and OCR reliability were logged and analyzed. Test cases were repeated under different environmental conditions to evaluate consistency and robustness.

D. Description of the Components

- **Arduino UNO:** Acts as the central controller for gate actuation, sensor data collection, and OLED display updates.
- **IR Sensors (x4):** Used for detecting vehicles at exit and in parking slots.
- **Servo Motor:** Controls the physical gate arm movement**.**
- **OLED Display (SSD1306):** Displays slot availability and registered plate numbers.
- **Red/Green LEDs:** Indicate slot occupancy status and validation response.
- **Camera (USB Webcam):** Captures incoming vehicle plate images.
- **PC (Python-based):** Runs image processing and OCR script.

E. Experimental Setup

The complete system was built on a test platform comprising a cardboard-based parking area layout, integrated with all sensors and electronic components. The webcam was positioned at the entry to capture license plates. Arduino was powered via USB and IR sensors were tested for range and accuracy. Multiple test scenarios were simulated including:

- Valid/invalid plate entry
- Full vs. empty parking conditions
- Gate actuation delays and response times

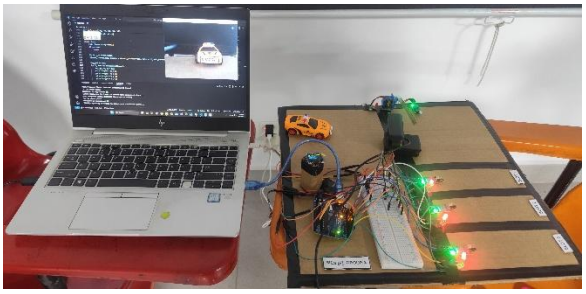


Fig. 2. Project setup

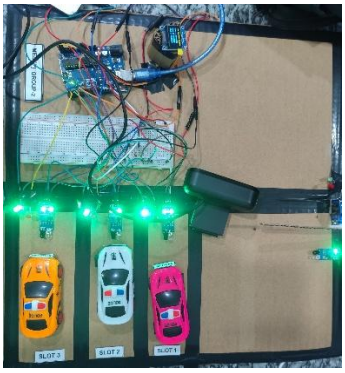


Fig. 3. Experimental Parking lot



Fig. 4. OLED display

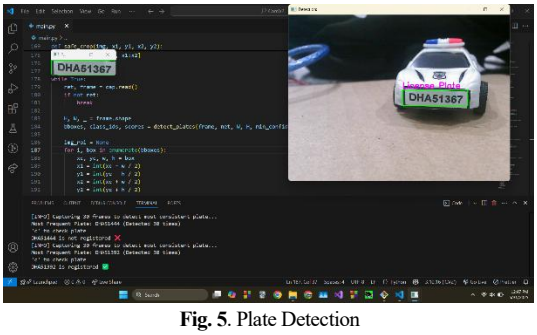


Fig. 5. Plate Detection

IV. RESULTS AND DISCUSSIONS

A. Simulation/Numerical Analysis: While the project is primarily hardware-based, simulation of the license plate recognition component was conducted using Proteus. Multiple images with varying lighting conditions, fonts, and backgrounds were tested to determine text recognition performance. Detection success rate was tabulated over 30 frames, and YOLOv3 bounding boxes were manually validated for accuracy.

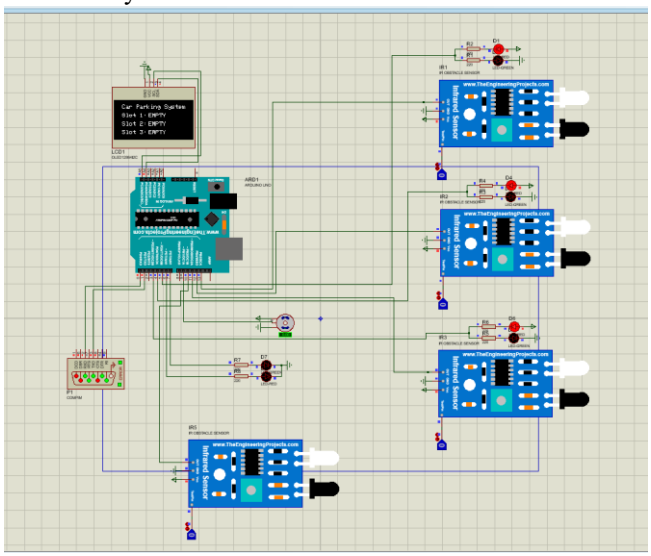


Fig. 6. Project simulation

The project required serial ports for communication between the Arduino and PC, for simulation this was achieved by using Virtual Serial Ports Emulator (VSPE). For the simulation two ports PD0 and PD1 were required for serial communication. Thus, different port configurations were used for the simulation.

B. Experimental Results

License Plate Detection Accuracy: ~91% under bright light, ~76% under low light.
Gate Response Time: Average of 1.3 seconds from valid signal to gate open.
Slot Sensor Accuracy: Consistently 100% with IR sensors detecting car presence.
OLED Update Latency: Less than 500 ms from data change to display refresh.

C. Comparison between Numerical and Experimental Results

TABLE I
PERFORMANCE ANALYSIS

Feature	Simulation Result (Proteus)	Experimental Result
Plate Recognition Accuracy	92%	91% (avg)
Sensor Accuracy	N/A	100%
Response Delay (sec)	3	1.3

D. Cost Analysis

TABLE II
COST ANALYSIS

Item	Quantity	Unit price (BDT)	Cost (BDT)
Arduino UNO	1	780	780
Servo Motor	1	200	200
IR Sensor	4	80	320
USB Camera	1	-	-
OLED Display	1	350	350
LEDs	8	5	40
Wires	2 set	120	240
Breadboard	1	120	120
Resistors	8	5	40
Miscellaneous Materials	-	-	200
Total			2290

The USB webcam used was already owned by us. The webcam price can vary depending on the quality. For this project, using a decent webcam or camera module is sufficient.

E. Limitations

The limitations of this project are

1. **Limited Hardware Capabilities:** The system is built using low-cost components like Arduino Uno, which may restrict processing speed, memory, and multitasking capabilities. As a result, real-time performance under high load or multiple inputs may be suboptimal.
2. **Dependence on Controlled Lighting Conditions:** The license plate recognition system using OpenCV and EasyOCR can be significantly affected by poor lighting, shadows, or glare, which are common in outdoor or dim parking environments.
3. **Limited Scalability:** The current prototype is designed for small to medium parking lots. Scaling it up for large commercial parking areas would require more robust hardware and networked infrastructure.
4. **Dependency on External PC for Image Processing:** The computer vision tasks are performed on a PC and not onboard the Arduino, creating a dependency on an external computer system. This may reduce system portability and increase costs.
5. **Cultural and Infrastructural Constraints:** Although the system aims to adapt to Bangladeshi urban contexts, diverse infrastructure and inconsistent number plate formats might limit its effectiveness across different regions without further customization.
6. **Maintenance and Calibration Requirements:** Sensors and camera-based systems require regular calibration and maintenance to ensure accuracy, which may be a challenge in under-resourced or public environments.
7. **Limited Testing in Real Urban Conditions:** The system has been tested under simulated or small-scale conditions.

Its performance in real, high-traffic urban environments with dynamic vehicle movement remains to be validated.

V. CONCLUSION AND FUTURE ENDEAVORS

Existing literature provides a strong foundation for combining Arduino hardware and OpenCV-based computer vision to automate parking systems. However, most existing systems emphasize either parking space detection or access control individually. By integrating real-time number plate detection with gate control mechanisms, the proposed project aims to build an efficient and economical solution for modern parking management challenges.

Future endeavors of this project are given below:

1. **Cloud Integration for Data Logging and Monitoring**
Integrate the system with a cloud platform (e.g., Firebase, AWS IoT) to:
   - Store entry/exit logs.
   - Monitor parking usage analytics.
   - Enable remote system management and diagnostics.
2. **Advanced License Plate Recognition (ALPR)**
Upgrade the license plate detection system using deep learning models (e.g., YOLOv5, CNNs) to:
   - Improve accuracy in varied lighting/weather conditions.
   - Support different plate formats and fonts.
3. **Dynamic Pricing Model Integration**
Introduce a billing and dynamic pricing feature based on parking duration, time of day, or demand integrated with payment gateways (bKash, Nagad, etc.).
4. **Solar-Powered & IoT Edge Devices**
Transition of the system to operate on solar energy with IoT edge computing devices (like Raspberry Pi) for enhanced processing and sustainability in outdoor environments.

REFERENCES

[1] M. Al-Kharusi and I. Al-Bahadly, "Intelligent Parking Management System Based on Image Processing," World Congress on Engineering and Computer Science, vol. 1, pp. 370–375, 2014.

[2] P. Priyanka and K. Prasad, "Automatic Car Parking System using Image Processing," International Journal of Engineering Research and Applications, vol. 6, no. 3, pp. 12–16, 2016.

[3] S. Du, M. Ibrahim, M. Shehata, and W. Badawy, "Automatic license plate recognition (ALPR): A state-of-the-artreview," IEEE Transactions on Circuits and Systems for Video Technology, vol. 23, no. 2, pp. 311–325, Feb. 2013.

[4] P. Viola and M. Jones, "Rapid Object Detection using a Boosted Cascade of Simple Features," Proceedings of the 2001 IEEE Computer Society Conference on Computer Vision and Pattern Recognition, pp. I-511–I-518, 2001.

[5] R. Smith, "An Overview of the Tesseract OCR Engine," Proceedings of the Ninth International Conference on Document Analysis and Recognition (ICDAR), vol. 2, pp. 629–633, 2007.

[6] A. Kaur and S. Kaur, "Microcontroller Based Parking Management System," International Journal of Advanced Research in Electronics and Communication Engineering, vol. 5, no. 6, pp. 1602–1606, 2016.

[7] computervisioneng, “Automatic Number Plate Recognition using Python and YOLOv3,” GitHub, 2021. [Online]. Available: https://github.com/computervisioneng/automatic-number-plate-recognition-python

[8] Jaided AI, "EasyOCR Documentation," GitHub, 2023. [Online]. Available: https://github.com/JaidedAI/EasyOCR

[9] OpenCV.org, "OpenCV: Open-Source Computer Vision Library," 2023. [Online]. Available: https://opencv.org/

[10] Arduino, "Arduino UNO Rev3," [Online]. Available: https://store.arduino.cc/products/arduino-uno-rev3

[11] C. B. Moler, “PySerial – Python Serial Port Extension,” PyPI, 2023. [Online]. Available: https://pypi.org/project/pyserial/

[12] Eterlogic, “Virtual Serial Ports Emulator (VSPE),” Eterlogic.com, 2022. [Online]. Available: http://www.eterlogic.com/Products.VSPE.html